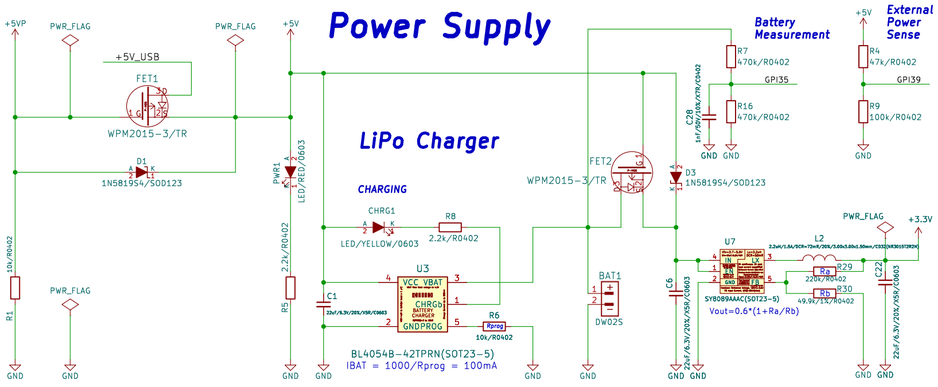
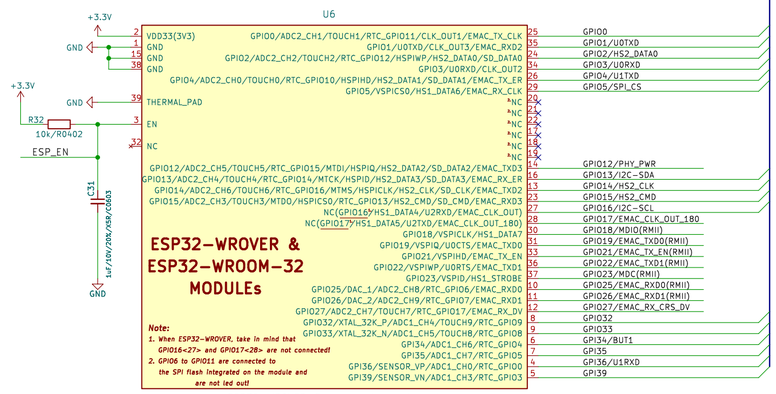
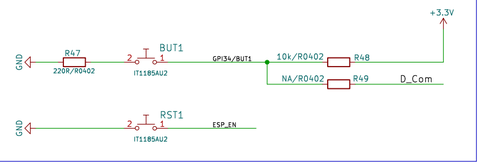
Power Supply



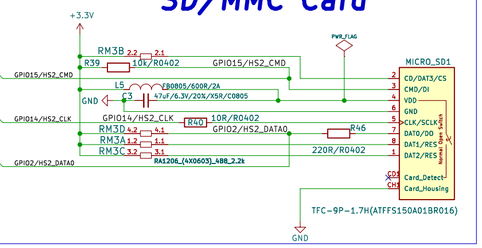
ESP32-WROOM-32E MODULE



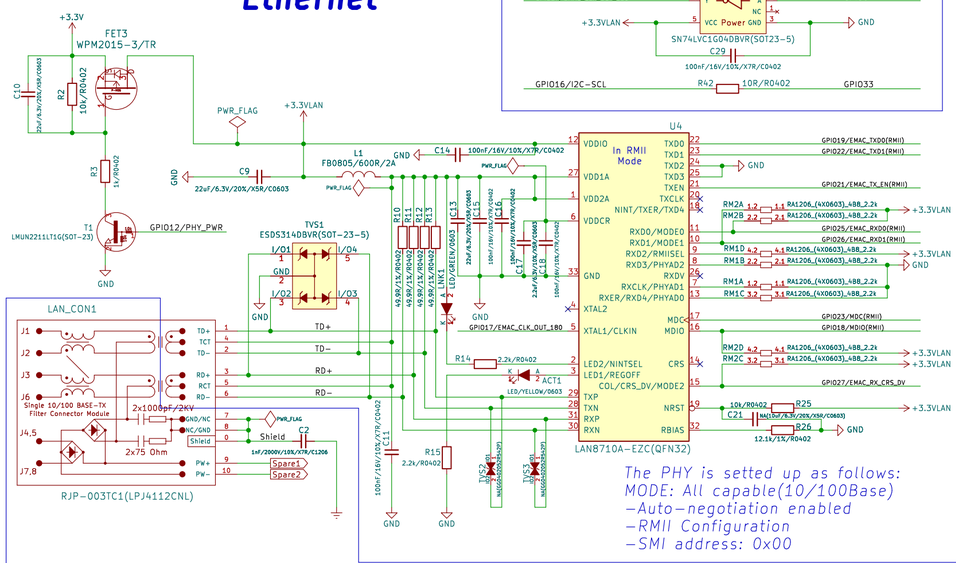
Buttons



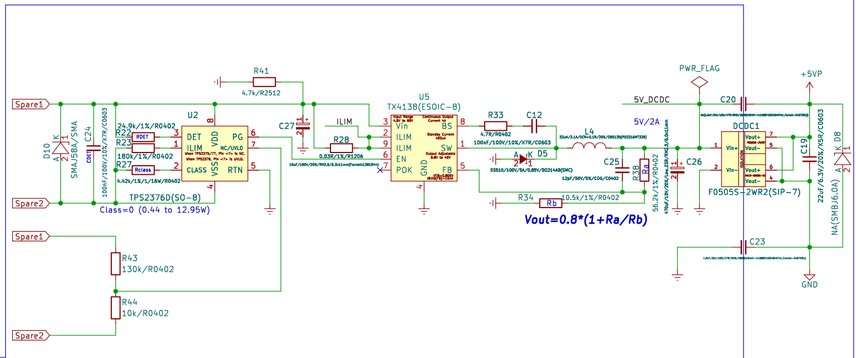
SD/MMC Card



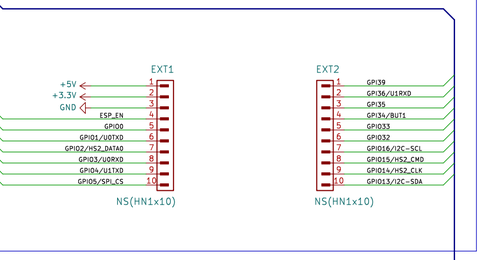
Ethernet



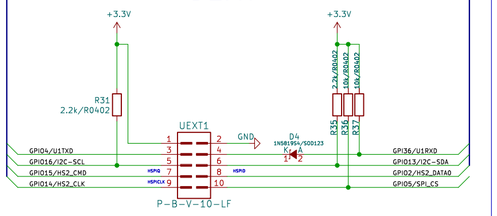
PoE Power Supply



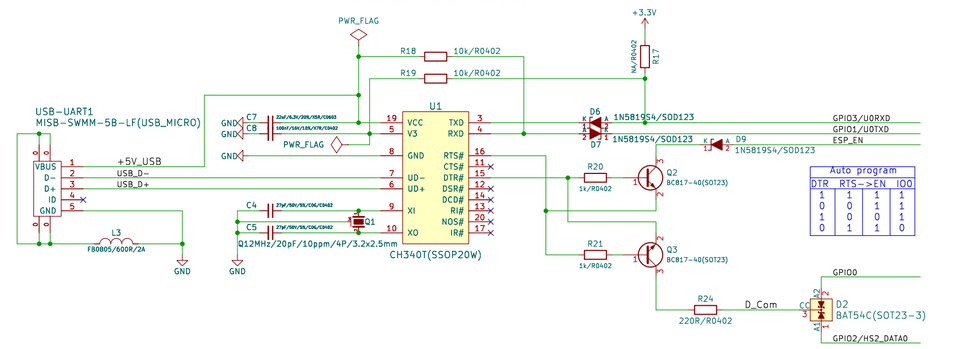
Extensions



UEXT



USB to UART



Notes:

Software Selectable Pins		
Interface	Signal	Pin
EMAC	EMAC_MDC_out EMAC_MDI_in EMAC_MDO_out EMAC_CRS_out EMAC_COL_out	Any GPIO
I2C	I2CEXT0_SCL_in I2CEXT0_SDA_in I2CEXT1_SCL_in I2CEXT1_SDA_in I2CEXT0_SCL_out I2CEXT0_SDA_out I2CEXT1_SCL_out I2CEXT1_SDA_out	Any GPIO
General Purpose SPI	HSPIQ_in/_out HSPID_in/_out HSPICLK_in/_out HSPI_CS0_in/_out HSPI_CS1_out HSPI_CS2_out VSPIQ_in/_out VSPID_in/_out VSPICLK_in/_out VSPI_CS0_in/_out VSPI_CS1_out VSPI_CS2_out	Any GPIO
For more information refer to esp_wroom_32_datasheet_en.pdf.		

Internal Bootstrapping Resistors	
MTDI/GPIO12:	Pull-Down
GPIO0:	Pull-Up
GPIO2:	Pull-Down
GPIO4:	Pull-Down
MTDO/GPIO15:	Pull-Up
GPIO5:	Pull-Up

Voltage of internal LDO(VDD_SDIO)					
Pin	Default	3.3V	1.8V		
GPIO12/MTDI	Pull-Down	0	1		
Booting Mode					
Pin	Default	SPI Flash Boot	Download Boot		
GPIO0	Pull-Up	1	0		
GPIO2	Pull-Down	Don't-care	0		
Debugging Log on U0TXD During Booting					
Pin	Default	U0TXD Toggling	U0TXD Silent		
GPIO15/MTDO	Pull-Up	1	0		
Timing of SDIO Slave					
Pin	Default	Falling-edge:Output	Falling-edge:Input Rising-edge:Output	Rising-edge:Input Falling-edge:Output	Rising-edge:Input Rising-edge:Output
GPIO15/MTDO	Pull-Up	0	0	1	1
GPIO5	Pull-Up	0	1	0	1

Bootstrapping Pins Settings

Fiducials

FID1 Fiducial
FID2 Fiducial
FID3 Fiducial
FID4 Fiducial
FID5 Fiducial
FID6 Fiducial

Mounting Holes

MH1 Mounting_Hole_NPTH
MH2 Mounting_Hole_NPTH
MH3 Mounting_Hole_NPTH

Galvanic Isolated Part

The PHY is setted up as follows:
MODE: All capable(10/100Base)
-Auto-negotiation enabled
-RMII Configuration
-SMI address: 0x00

OLIMEX
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open source hardware

Sheet: /
File: working.kicad_sch

Title: ESP32-PoE-ISO

Size: A2	Date: 2022-10-10	Rev: L
KiCad E.D.A. eeschema 7.0.6		Id: 1/1